Cairo University
Faculty of science
Mathematics Department
120 minutes.

Final Exam (Spring 2016)
Artificial Intelligence
Comp 408

Answer the following questions. The total credit is 60. **The Exam is in seven Pages**

Question (1) (17 points)

A. **(3 points)** Define an agent. Is any agent rational? What does it mean to be autonomous agent?

B. **(4 points)** The Towers of Hanoi is the following problem:
There are three sticks A,B, and C. And there three disks on stick A. The top has diameter of 1, the middle disk has a diameter of 2, and the bottom 3. No disks on sticks B and C. You should move them all in the same order to C. You must move one disk at time and you cannot place a larger disk on a smaller one.

i. Find a representation of the problem that can be easily programmed.

ii. Use your representation in (i) to describe the initial state , the goal state, and the successor function

P.T.O.

C. **(10 points)** Consider a state space where the **initial** state is number 1 and the successor function for state n returns two states, numbers 2n and 2n + 1. (Ex. The successors of state 1 are states 2 and 3)

i. Draw the portion of the state space for states 1 to 15.

ii. Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search.

iii. Would bidirectional search be appropriate for this problem? If so, describe in detail how it would work.

iv. What is the branching factor in each direction of the bidirectional search?

v. Does the answer to (iii) suggest a reformulation of the problem that would allow you to solve the problem of getting from state 1 to a given goal state with almost no search?

P.T.O.

Question (2) (15 points)

A. **(2 points)** Show that the following sentence in propositional logic is valid.
$(p \rightarrow Q) \leftrightarrow (\sim Q \rightarrow \sim P)$

B. **(8 points)** Circle T or F for True/False: **(two circles is a wrong answer)**

i. (T / F) Iterative Deeping Search Algorithm is optimal.
ii. (T / F) An agent is usually operating in a partially observable environment.
iii. (T / F) A* Search cannot be used in robotics because percepts, states, and actions are continuous.
iv. (T / F) Depth-first Iterative deepening always returns the same solution as breadth-first search if b is finite.
v. (T / F) The clause $(\sim A \vee \sim B \vee C)$ entails the clause $(\sim A \vee \sim B)$
vi. (T / F) A Sound inference rule can derive all sound sentences from a knowledge base
vii. (T / F) Context-free grammars may fail to capture English grammar because the meaning of an English sentence may depend on the context of the utterance.
viii. (T / F) Ambiguity in natural language can always be resolved by checking dictionary definitions of words

C. **(3 points)** What affects the speed of the genetic algorithm?

D. **(2 points)** Is Forward chaining complete in first-order logic? Explain why?

P.T.O.

Question (3) (15 points)

i. **(7 points)** Represent the following knowledge base in first-order logic. Use the predicates: attend(x), fair(t), pass(x,t), prepared(x), smart(x), study(x), cu_std(x).

a. Everyone who is smart, studies, and attends class will be prepared.
b. Everyone who is prepared will pass a test if the test is fair.
c. Every Cairo University student i.e. cu_std is smart.
d. If a test is not fair, no one will pass the test
e. Hany is a cu_student
f. At least one student passed the AI test
g. Hany attends class.

ii. **(4 points)** Convert this knowledge base into conjunction normal form, CNF.

P.T.O.

iii. **(4 points)** Use resolution to prove that:

Study(Hany) → pass (Hany, AI)

P.T.O.

Question (4) (13 points)

A. **(7 points)** Consider the game of 2×2 tic tac toe (X-O) where each player has the additional option of passing (i.e. making no action). Assume O (The player Min) goes first.

i. Draw the full game tree down to depth 2. (first Min's Turn, then Max's Turn)
ii. Suppose the evaluation function is the number of Xs minus the number of Os. Mark the values of all leaves Then propagate upward using Minimax fashion.
iii. Circle any node that would not be evaluated by alpha-beta pruning.
iv. Suppose we want to solve the game to find the optimal move. Explain why apha-beta would be better than minimax.

P.T.O.

B. **(6 points)** Consider the sentence "the man walked slowly to the supermarket" and consider the following lexicon and grammars:

Noun –> man | supermarket
V –> walked
Adv –> slowly
Prep –> to
Det –> the

A

S	–>	NP VP
NP	–>	Noun
NP	–>	Article NP
VP	–>	Verb Vmod
Vmod	–>	Adv Vmod
Vmod	–>	Adv
Adv	–>	PP
PP	–>	Prep NP

B

S	–>	NP VP
NP	–>	Noun
NP	–>	Article NP
VP	–>	Verb Adv
Adv	–>	Adv Adv
Adv	–>	PP
PP	–>	Prep NP

C

S	–>	NP VP
NP	–>	Noun
NP	–>	Article Noun
VP	–>	VP PP
VP	–>	VP Adv Adv
VP	–>	Verb
PP	–>	Prep NP

i. Which of the three grammars, combined with the lexicon, generate the given sentence.
ii. Show the corresponding parse tree.
iii. Write down one sentence of <u>non-English</u> generated by one of the above grammars. (Explain how it is generated by the grammar)

Best Wishes
Areeg Abdalla